Recommendation Scores

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected eligibility-and-scoring record.

Calculate a score, check the boundary, reject ineligible winners and compare preferences.

Keep one habit: Retain field meanings, hard requirements, weights, raw components, ties and human decisions.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Nearest Neighbor Reasoning

Repair Bias in Data

Noise and Outliers

W1 · Score an already eligible activity

> Use the supplied fun/learning roles.

Your evidence record

- Opening activity rule: fun/learning ratings0–5, score2*fun+learning, duration<=20. The separate pictured interest/practice case has limit14. Main workbook W1 supplies eligibility; W2 uses the20-minute boundary; W5 explicitly supplies two already-eligible options with its own ratings. Independent W4 has a fresh15-minute limit and records. Do not transfer an old case's names, limits or scores into a new one.

- An eligible activity has fun 2 and learning 4. Use 2 * fun + learning.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Check the inclusive20-minute boundary

> At most includes equality.

Your evidence record

- Opening activity rule: fun/learning ratings0–5, score2*fun+learning, duration<=20. The separate pictured interest/practice case has limit14. Main workbook W1 supplies eligibility; W2 uses the20-minute boundary; W5 explicitly supplies two already-eligible options with its own ratings. Independent W4 has a fresh15-minute limit and records. Do not transfer an old case's names, limits or scores into a new one.

- Which durations 19, 20, 21 qualify under at most 20?

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Reject an ineligible high-scoring option

> A larger preference score cannot override the requirement.

Your evidence record

- Opening activity rule: fun/learning ratings0–5, score2*fun+learning, duration<=20. The separate pictured interest/practice case has limit14. Main workbook W1 supplies eligibility; W2 uses the20-minute boundary; W5 explicitly supplies two already-eligible options with its own ratings. Independent W4 has a fresh15-minute limit and records. Do not transfer an old case's names, limits or scores into a new one.

- An ineligible item **score**s 15 and an eligible item **score**s 9. Which should be offered under the stated policy?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Compare two explicit preference rules

> Keep eligible data fixed while changing only weights.

Your evidence record

- Opening activity rule: fun/learning ratings0–5, score2*fun+learning, duration<=20. The separate pictured interest/practice case has limit14. Main workbook W1 supplies eligibility; W2 uses the20-minute boundary; W5 explicitly supplies two already-eligible options with its own ratings. Independent W4 has a fresh15-minute limit and records. Do not transfer an old case's names, limits or scores into a new one.

- For eligible A (fun 4, learning 1) and B (fun 2, learning 4), compare **weight**s (2,1) with (1,2).

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Choose independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate **score**s with 2 * fun + learning and choose.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Choose independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Publish the features, **weight**s, raw **score**s, and reason for the top result.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Check a boundary for **Weight**ed **score**s and **constraint**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Weight**ed **score**s and **constraint**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Score=3×genreMatch+2×short+illustrated. Compute A=[1,0,1], B=[0,1,1], C=[1,1,0]. A reader now requires illustrations. Which book wins after applying that **constraint**?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Score=2 times genre match + length match. A matches both, B genre only, C length only.

Score and **rank**.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A recommender gives 3 points for topic match and 1 for difficulty match. Book A matches topic only; B matches difficulty only. Compute **score**s and justify the **rank**ing.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 23 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Score=3×genreMatch+2×short+illustrated. Compute A=[1,0,1], B=[0,1,1], C=[1,1,0]. A reader now requires illustrations. Which book wins after applying that **constraint**?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate **score**s with 2 * fun + learning and choose.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional book-activity chooser has a 14-minute limit.

Score=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

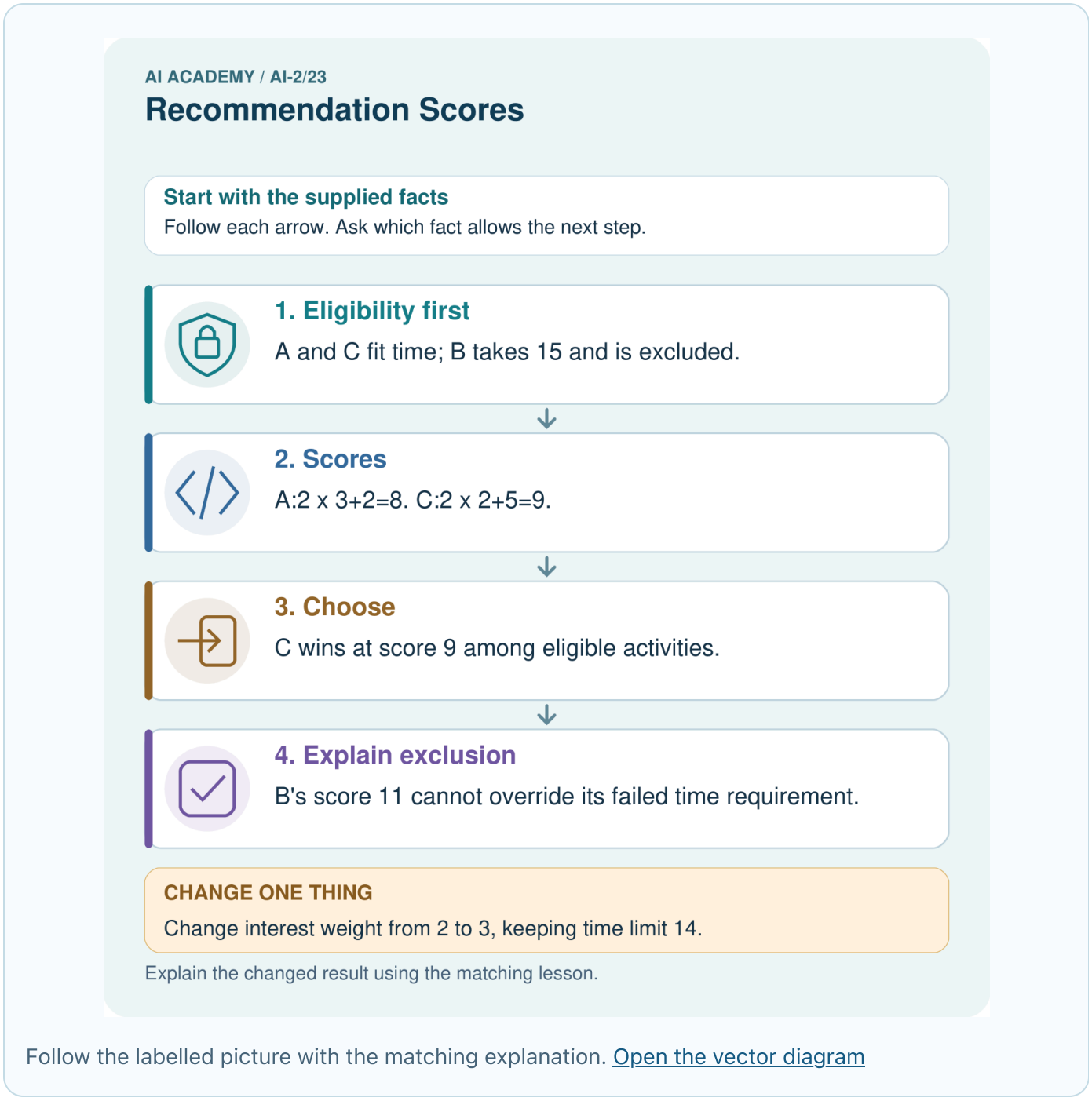
AI ACADEMY / AI-2/23

Time filtering and scores

	Minutes	Score	Status
A	14	8	eligible
B	15	11	excluded
C	12	9	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why does B not become the winner when its **score** is largest?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional book-activity chooser has a 14-minute limit.
- Which labelled step matches this detail? B's **score** 11 cannot override its failed time requirement.
- 1. Choose
- 2. Explain exclusion
- 3. Eligibility first
- 4. **Score**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Explain exclusion B's **score** 11 cannot override its failed time requirement.

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Changed-case question

- Changed-case practice: Change interest **weight** from 2 to 3, keeping time limit 14. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. A **score**s 11; C also **score**s 11; tie-by-ID chooses A.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: A **score**s 11; C also **score**s 11; tie-by-ID chooses A. B remains too long; changing preferences changes only **rank**ing of A and C.

Repair and save your work

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and **rank training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

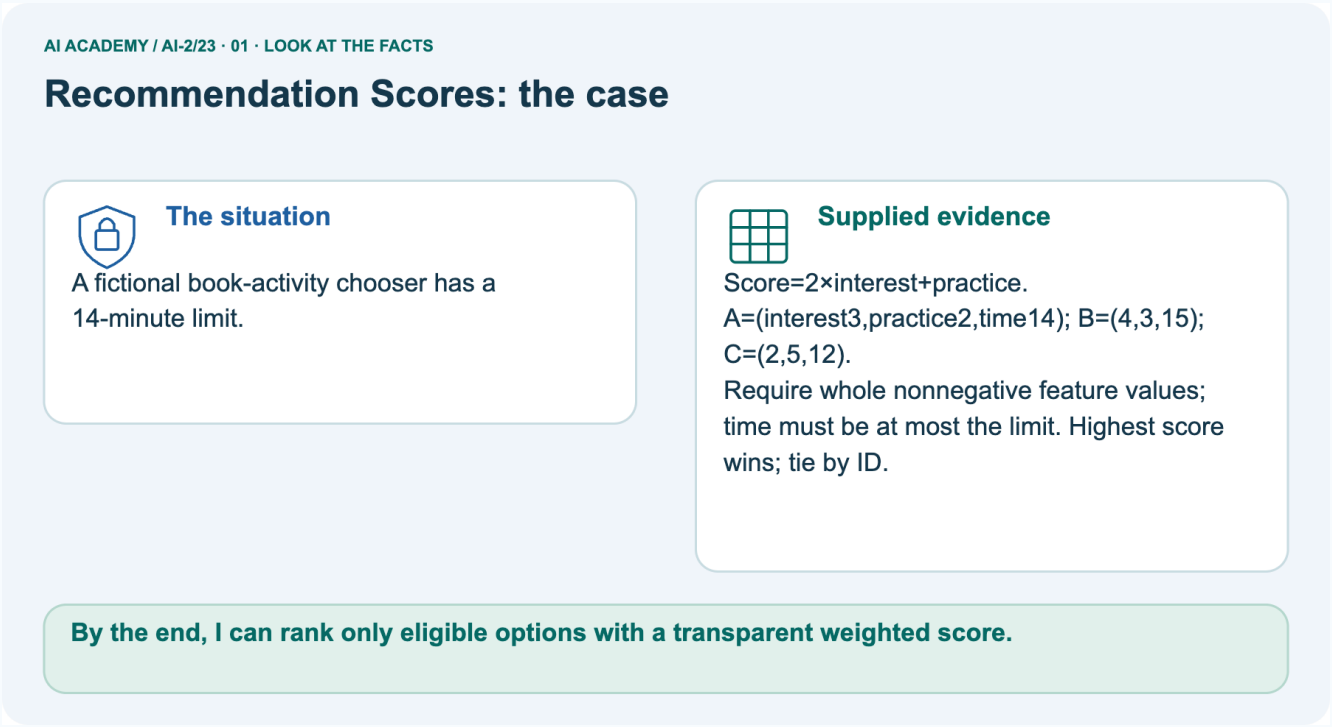
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Recommendation Scores: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

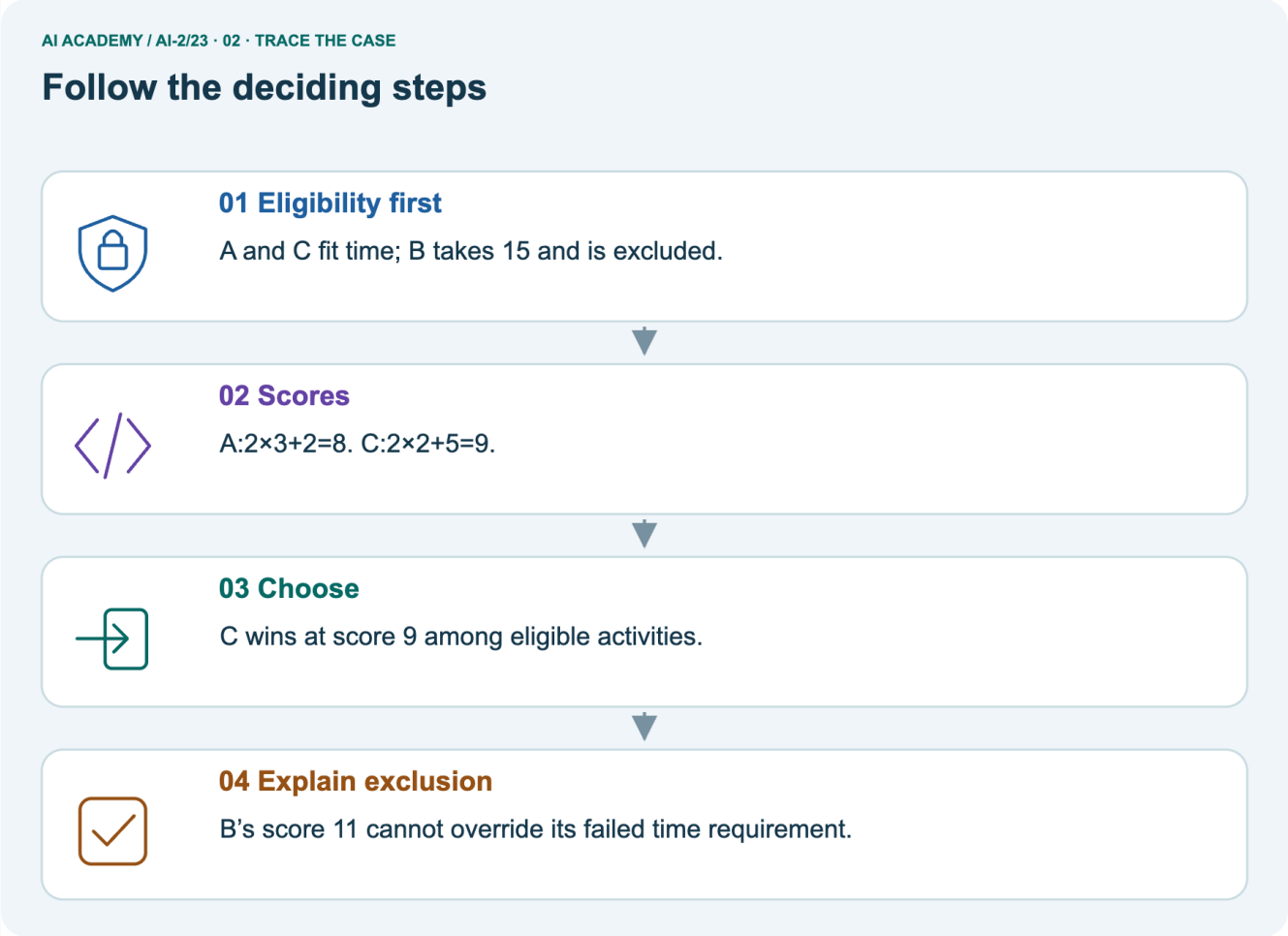
A fictional book-activity chooser has a 14-minute limit. Supplied facts: **Score**=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

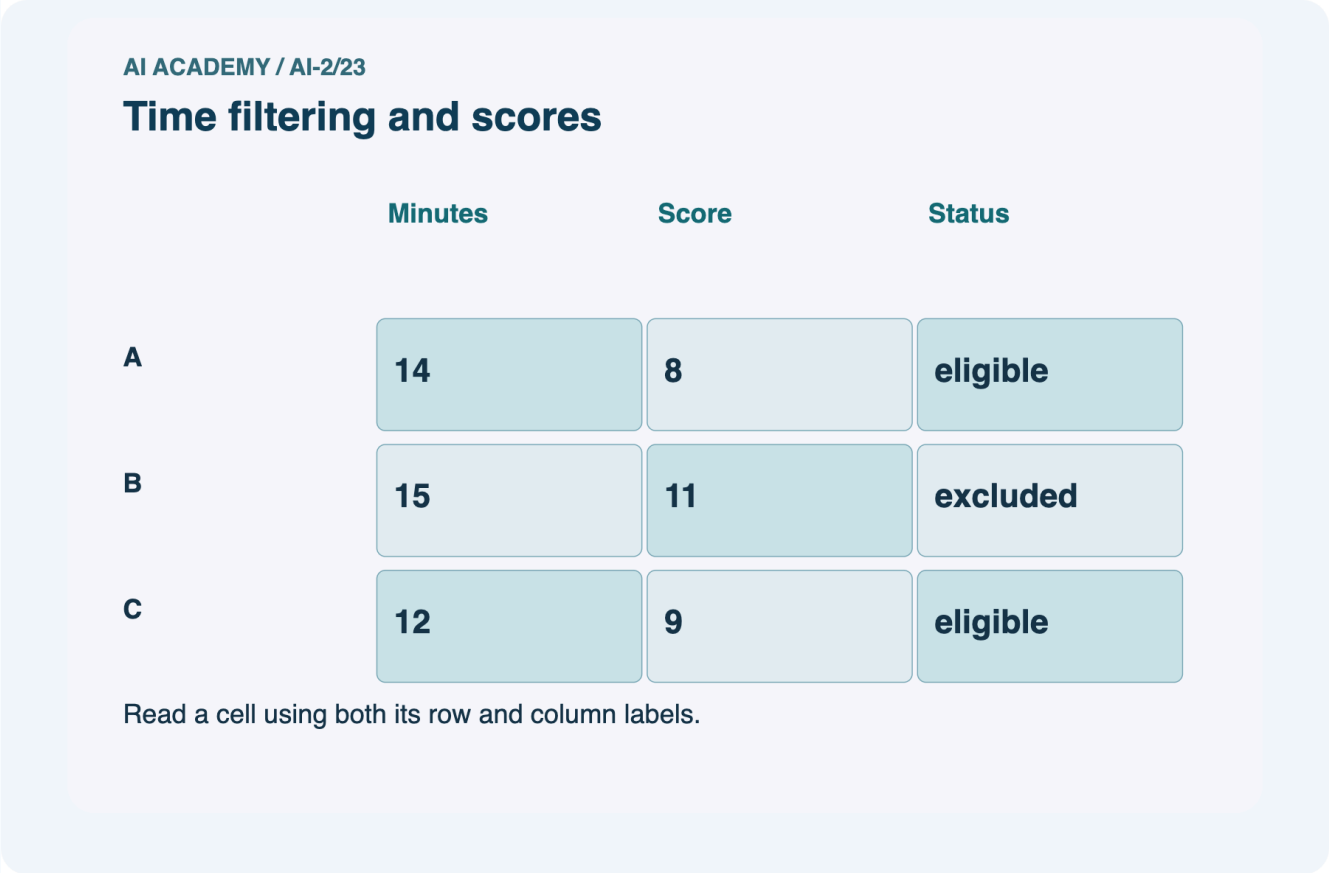
Eligibility first: A and C fit time; B takes 15 and is excluded. → **Score**s: A:2×3+2=8.

C:2×2+5=9. → Choose: C wins at **score** 9 among eligible activities. → Explain exclusion: B's **score** 11 cannot override its failed time requirement.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Time filtering and scores



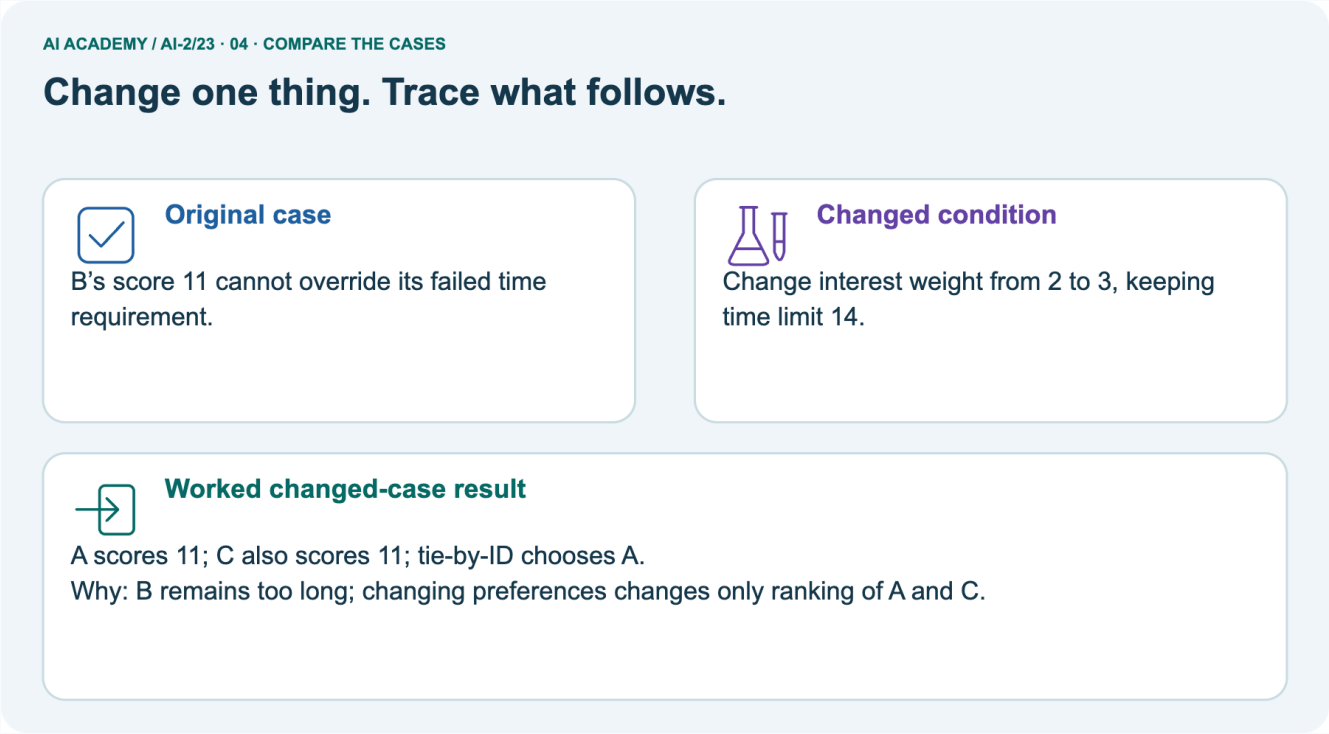
Follow the labelled picture with the matching explanation. Open the vector diagram

Time filtering and **score**s. A: Minutes = 14, **Score** = 8, Status = eligible. B: Minutes = 15, **Score** = 11, Status = excluded. C: Minutes = 12, **Score** = 9, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

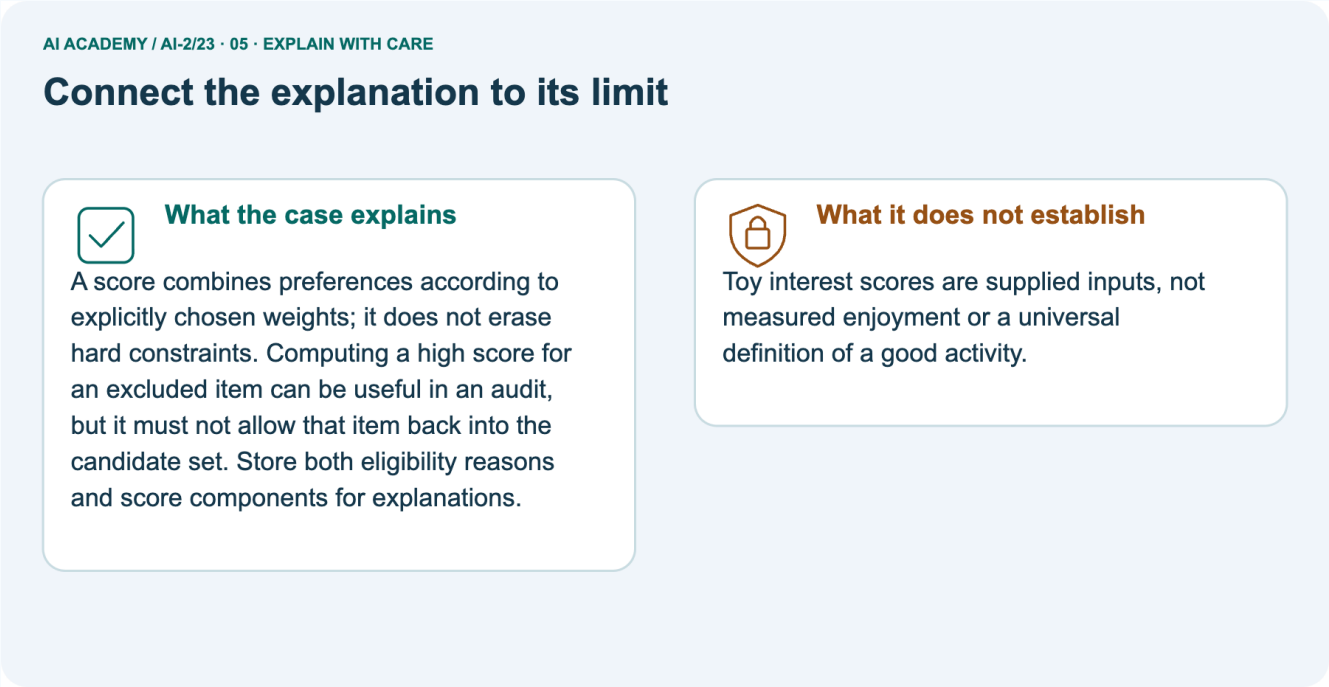
Original case: B's **score** 11 cannot override its failed time requirement. Changed condition: Change interest **weight** from 2 to 3, keeping time limit 14. Expected result: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

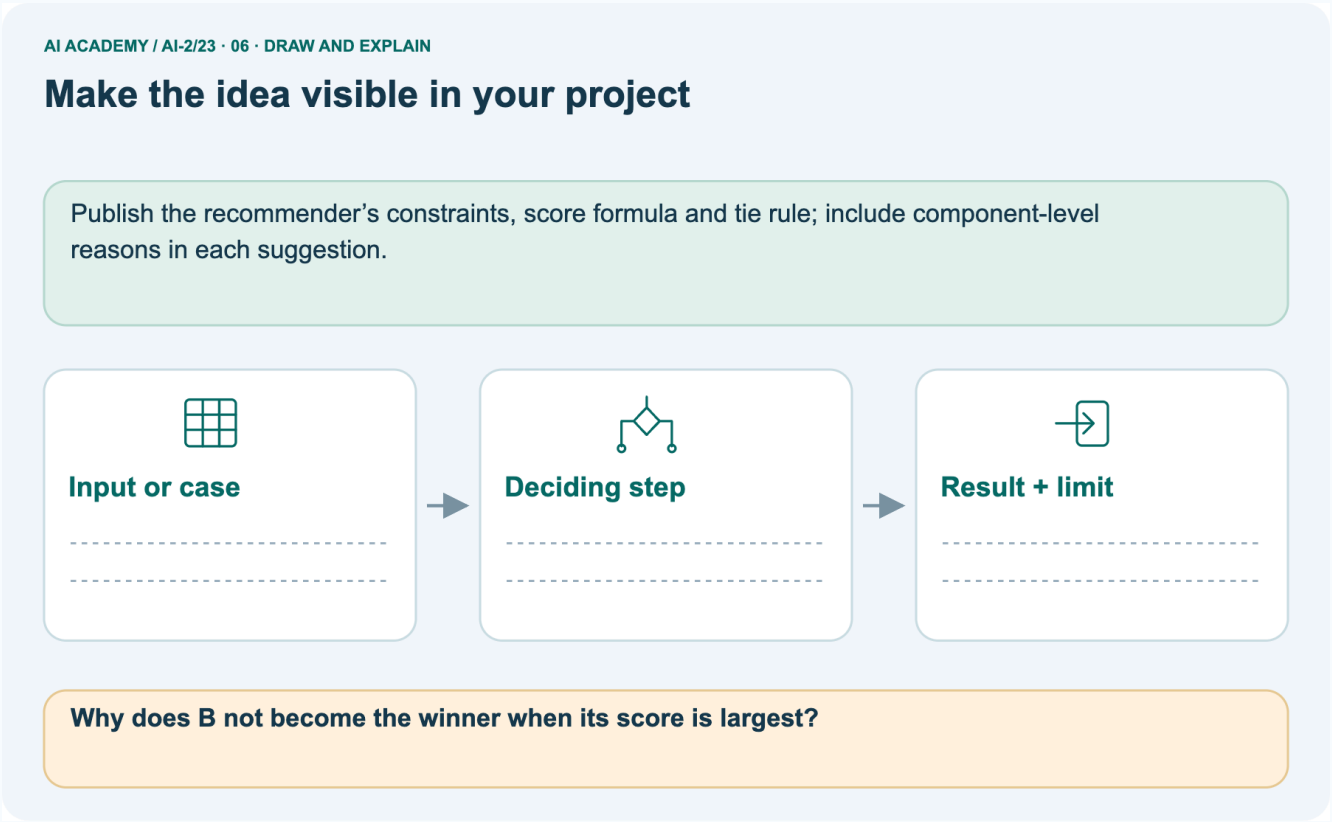
Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Limit: Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion. Draw your **input**, deciding step, result and limit.

Supported question: Why does B not become the winner when its **score** is largest?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Recommendation Scores: the case

02. Follow the deciding steps

03. Time filtering and scores

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.